PHASE 1 PROJECT REPORT

Electronic Design, Embedded Programming and 3D Enclosure Development

Project Title: Smart Automated Prototype System
Phase: Phase 1
Prepared by: Project Team
Location: Bengaluru, Karnataka, India
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This report documents the completed Phase 1 activities, including system planning, electronic design, embedded programming, prototype demonstration, CAD enclosure development, testing documentation, and the project journey. Physical validation remains part of the next implementation stage unless explicitly identified as completed.

Abstract

This Phase 1 project presents the design and documentation of a smart automated prototype system integrating electronic control, embedded programming, mechanical enclosure development, and a planned actuation mechanism. The work establishes the foundation for a functional prototype through system-level planning, component selection, pin mapping, Arduino-based control logic, and 3D CAD development of the enclosure, box, lid, servo mount, and linkage arrangement. A dummy hardware setup and demonstration materials were prepared to communicate the intended operation. Testing documentation has been organized as simulated or planned validation; no physical performance values are claimed until measurements are obtained from the assembled prototype. The outcome of Phase 1 is a coherent, review-ready design package that can be implemented and physically validated during Phase 2.

1. Introduction

The project addresses the need for a compact, low-cost, and understandable automated system that can be controlled using a microcontroller and a servo-based mechanism. The design combines electronics, software, and mechanical packaging rather than treating each subsystem independently. This approach makes it possible to verify the system architecture early and reduce integration problems during physical assembly.

1.1 Problem Statement

Many small automation prototypes demonstrate control logic without providing a suitable enclosure, clear mechanical arrangement, or structured validation plan. Such prototypes may be difficult to assemble, explain, transport, or improve. This project therefore focuses on developing a documented and mechanically organized prototype platform with a clearly defined Phase 1 scope.

1.2 Phase 1 Scope

The Phase 1 scope is limited to electronic design, embedded programming, and 3D enclosure development. Advanced communication, IoT connectivity, cloud integration, and ESP32-based extensions are reserved for Phase 2 and are not represented as completed Phase 1 features.

2. Objectives

The main objectives are to:

• Define the system architecture and operating sequence.

• Select and document the required electronic and mechanical components.

• Develop the microcontroller control logic and software flow.

• Prepare pin connections and an implementation-ready wiring plan.

• Design a compact 3D enclosure with a box, lid, servo mounting arrangement, and planned linkage.

• Prepare testing documentation without presenting unmeasured values as experimental results.

• Create an organized resource structure for reports, code, images, and CAD models.

3. System Overview and Working Principle

The system operates through a microcontroller that receives an input or command, evaluates the programmed logic, and generates an output signal for the actuator. The servo converts the electrical control signal into angular motion. The mechanical linkage transfers this motion to the intended moving part of the enclosure or demonstration mechanism. Power, signal routing, and mechanical clearance are considered during the design so that the electrical and physical subsystems can be integrated in Phase 2.

3.1 Functional Flow

Stage	Function
1. Input/command	Receives the user or programmed trigger.
2. Processing	Microcontroller executes the control logic.

3. Actuation	Servo receives a position command.
4. Mechanical transfer	Linkage converts servo movement into the required motion.
5. Feedback/observation	System response is observed during demonstration and later physical testing.

4. Hardware and Electronic Design

The electronic subsystem is designed around an Arduino-compatible microcontroller platform, a servo actuator, a regulated power source, connecting wires, and the required supporting components. The exact component values and final wiring must be verified against the selected hardware before assembly. The design emphasizes simple control, accessible programming, and ease of troubleshooting.

4.1 Component Summary

Subsystem	Component/role
Controller	Arduino-compatible microcontroller for program execution.
Actuator	Servo motor for controlled angular movement.
Power	Suitable regulated supply selected according to controller and servo requirements.
Interconnection	Jumper wires, connectors, breadboard or PCB wiring as applicable.
Mechanical interface	Servo horn, linkage, mounting features, box, and lid.

4.2 Pin Connections

The pin-connection table from the original Phase 1 report is retained as the design reference. Before powering the prototype, each connection should be checked for signal compatibility, common ground, supply voltage, and accidental short circuits. Servo power should be evaluated separately from the controller supply when the actuator current demand requires it.

5. Embedded Software

The Arduino software implements initialization, input handling, actuator control, and repeated operation. The program is structured to keep hardware control separate from decision logic, allowing later changes to the mechanism without rewriting the complete application.

5.1 Software Sequence

• Initialize the controller pins and servo interface.

• Set the actuator to a safe starting position.

• Read the defined input or command.

• Apply the programmed decision logic.

• Move the servo to the required position using the defined timing.

• Return to the monitoring state or repeat the operating cycle.

The final source code should be included in the project resource package under the code directory. Any future revisions should be versioned and identified with a date or release label.

6. 3D CAD and Mechanical Development

The CAD work converts the electronic concept into a practical physical form. The completed design package includes the latest enclosure concept, final render, updated box and lid, servo mounting arrangement, and the planned servo/linkage mechanism. The enclosure is intended to protect the electronics, provide a repeatable mounting location, and make the prototype easier to demonstrate and modify.

6.1 Design Considerations

• Adequate internal clearance for the controller, wiring, and actuator interface.

• A removable or accessible lid for assembly, inspection, and maintenance.

• Servo alignment with the linkage to reduce binding and unintended loads.

• Clearance for the moving parts across the complete range of motion.

• Mounting features that can be revised after physical fit testing.

• A clean external form suitable for demonstration and documentation.

The final CAD screenshots and rendered views should be placed in the images directory, while editable model files should be stored in the models directory. The CAD design is complete as a documentation deliverable; dimensional and mechanical performance claims remain subject to physical verification.

7. Prototype Demonstration and Testing

A dummy hardware setup and demonstration video were prepared to communicate the proposed operation and user interaction. These materials demonstrate the concept and presentation flow; they must not be interpreted as measured results from a fully assembled physical prototype.

7.1 Testing Documentation

The file *06_Testing/Test_Results.pdf* is referenced as the project testing document. At the current stage, it is classified as simulated/planned testing documentation. It may describe expected operation, test cases, acceptance criteria, and observations from the demonstration, but numerical performance values should only be added after physical measurements are recorded.

Test area	Phase 1 status	Required Phase 2 evidence
Control logic	Prepared and demonstrated conceptually.	Verified response on assembled hardware.
Servo movement	Planned/observed in demonstration context.	Measured angular range, repeatability, and load response.
Enclosure fit	Completed in CAD.	Physical fit, fastening, and clearance inspection.
Power and wiring	Designed at system level.	Voltage/current measurements and thermal observation.
Overall operation	Prototype demonstration material available.	End-to-end test with real hardware and recorded results.

7.2 Measurement Policy

No physical accuracy, speed, current, range, reliability, or success-rate claim is made in this report without a corresponding measurement record. This distinction preserves technical credibility and provides a clear starting point for Phase 2 validation.

8. Project Journey and Team Contributions

The project journey records the progression from problem identification and system planning to electronic design, embedded software preparation, CAD development, documentation, and demonstration. The team contributions should be maintained as a separate journey file and included in the final submission package.

Recommended contribution categories include:

• System architecture and technical planning.

• Circuit, component, and pin-connection documentation.

• Arduino programming and control-flow development.

• 3D modeling, enclosure refinement, rendering, and linkage planning.

• Testing documentation, media preparation, and report compilation.

9. Limitations and Risks

• Physical assembly and end-to-end testing are not represented as complete unless supported by measurement records.

• Servo torque, linkage strength, enclosure tolerances, and power integrity require physical verification.

• The dummy setup may not reproduce the electrical, mechanical, or environmental behavior of the final prototype.

• CAD dimensions may require revision after fabrication and component fit checks.

• Future connectivity and IoT features are outside the Phase 1 scope.

Risk mitigation will include a staged power-up procedure, continuity checks, mechanical free-movement checks, current measurement, and incremental software testing before full operation.

10. Phase 2 Future Scope

Phase 2 will convert the documented design into a physically validated prototype. Planned activities include fabrication or 3D printing of the enclosure, installation of the electronics, servo and linkage integration, power verification, calibration, functional testing, measurement recording, and design iteration. If required by the project goals, ESP32 or IoT capabilities may be evaluated in Phase 2 after the core local-control prototype is stable.

11. Resource Structure

Directory	Contents
documents/	Phase reports, testing documents, journey file, and references.
code/	Arduino source code, libraries, configuration notes, and revisions.
images/	CAD screenshots, final renders, wiring images, and demonstration photographs.
models/	Editable CAD files, exported meshes, and mechanical drawings.
media/	Prototype demonstration video and presentation assets.

12. Conclusion

Phase 1 successfully establishes the technical and documentation foundation for the smart automated prototype. The project includes a defined scope, system flow, hardware plan, embedded software approach, completed CAD enclosure development, demonstration materials, and a structured testing and resource plan. The report deliberately separates design completion and conceptual demonstration from physical measurement results. This makes the document technically accurate while providing a clear implementation path for Phase 2, where the prototype will be assembled, measured, evaluated, and improved.